

Bill of Materials - VBF-T3R2-BOM-01 (PDF release of bom.xlsx)

Item	Material	Thickness (um)	Volume fraction	Density (kg/m3)	Mass (g/cell)	Mass (kg/kWh)	Source / formula note
Positive active material	NMC811	45	0.528	3262	7.959775824000001	0.6564502964084915	layer mass x 96% literature AM split (parameter set has no binder/additive fractions)
Positive conductive additive	carbon black (lit. default)	45	0.011000000000000001		0.16582866300000004	0.013676047841843574	layer mass x 2% literature default
Positive binder	PVDF (lit. default)	45	0.011000000000000001		0.16582866300000004	0.013676047841843574	layer mass x 2% literature default
Negative active material	graphite	56	0.579	1657	5.5177185336	0.45505150483749934	layer mass x 96.5% literature AM split
Negative conductive additive	carbon black (lit. default)	56	0.006		0.0571784304	0.004715559635621755	layer mass x 1% literature default
Negative binder	CMC/SBR (lit. default)	56	0.015		0.142946076	0.011788899089054387	layer mass x 2.5% literature default
Separator	polyolefin (parameter-set values)	12			0.25930928400000003	0.02138548372556198	calc-energy layer mass (thickness x (1-porosity) x density)
Electrolyte	1 M LiPF6-class, high-transport (lit. density 1.2 g/cm3)			1200	5.951259600000001	0.49080604966845115	pore volume (eps-based) x literature density 1.2 g/cm3
Positive current collector	Al	16		2700	4.436640000000001	0.3658939281023864	parameter set thickness x density x area
Negative current collector	Cu	12		8960	11.042304000000001	0.9106693321659395	parameter set thickness x density x area
Enclosure, tabs, seals	Not modeled						beyond pure-simulation boundary
Total (contract caliber, electrolyte excluded)					29.75	2.453	cell/r2_p1_energy.json:mass_kg
Grand total including electrolyte (informative)					35.7	2.944	contract mass + electrolyte (lit. density)